

PREDICTION ALONE IS NOT PLANNING: GEOMETRIC ADVANTAGE RANKING FOR CONTROLLED LANGUAGE REASONING

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ABSTRACT

Agents acting through language must compare consequences rather than merely generate plausible text. We study action-conditioned joint-embedding prediction in a controlled language planning environment. Controlled-future equivalence defines a state sufficient for decision making, yet accurate transition prediction leaves its metric underdetermined. Two latent coordinate systems can encode identical transitions and reverse the ordering of actions by goal distance. We introduce Geometric Advantage Ranking (GAR), which learns same-state order constraints from observed counterfactual consequences. GAR may supervise a scorer over predicted successors or an information-matched direct scorer. On stylized arithmetic dependency puzzles, predictive GAR solves 43.3% of episodes in the minimum number of actions across five seeds, compared with 18.3% for latent prediction alone. The strongest sentence model reaches 52.1%. Five-seed ablations attribute the larger gain to counterfactual ordering and a further gain to successor prediction. A local audit finds substantially better action ordering from the learned GAR energy than from raw latent distance, while global effective rank does not track planning. In this environment, predictive state is a viable substrate for reasoning only when training also identifies how action consequences should be ordered.

1 INTRODUCTION

Language models are trained to predict text, even when the task is to choose an action according to what it will cause. Generation handles this problem indirectly. A model assigns probability to candidate words or sentences, and those probabilities become action scores. We ask whether consequence prediction offers a different route: encode the current reasoning state, predict how each candidate phrase changes it, and choose among the predicted outcomes.

Joint-embedding predictive architectures make that decomposition explicit (LeCun, 2022; Assran et al., 2023). They predict a target representation rather than reconstructing every target token. Action-conditioned variants can serve as latent world models and have supported planning in physical environments (Assran et al., 2025). Their use for reasoning raises a basic issue. An accurate world model specifies which state follows an action, but a planner also needs to know which successor is preferable.

Transition prediction does not settle that preference. Any invertible change of latent coordinates can preserve the learned dynamics. Unless the change also preserves the planning metric, it can reverse which successor appears closer to a goal. We study the resulting gap between predictive error and planning error.

We introduce Geometric Advantage Ranking (GAR), an objective that compares factual and counterfactual consequences available during training. GAR imposes a local order among actions considered at the same history. Our main architecture scores an action through its predicted successor. A matched direct-GAR control uses the same order supervision while bypassing the successor in the policy path. Their comparison asks whether predictive factorization contributes beyond supervised candidate ranking.

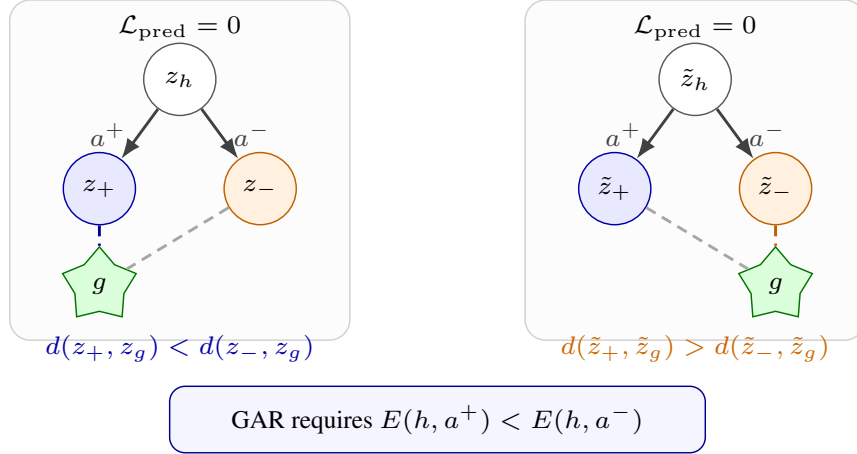


Figure 1: Both latent systems encode the same action-conditioned transitions. The left metric ranks a^+ closer to the goal. A valid re-embedding reverses that order on the right. Transition prediction alone cannot choose between them. GAR supplies the missing order constraint.

The experiments use stylized arithmetic dependency problems. Every method receives the same shuffled set of currently executable intent phrases. The environment executes the chosen phrase and returns its textual consequence. Fixing the candidate interface removes proposal recall and syntax validity from the comparison. The study therefore concerns conditional planning among feasible language actions, not unrestricted language generation.

Four questions organize the empirical study. RQ1 asks whether an action-conditioned JEPA can plan under the common interface. RQ2 asks whether latent transition prediction is sufficient. RQ3 asks whether GAR produces local order structure that predicts closed-loop success. RQ4 asks what JEPA representations preserve, discard, and expose relative to generative baselines.

The MLP JEPA reaches 43.3% strict success across five seeds. The token LM reaches 42.2%, the sentence LM reaches 47.1%, and a sentence model with auxiliary latent prediction reaches 52.1%. The best strict result therefore belongs to the sentence-plus-latent model. The objective controls address a different question. Factual and counterfactual latent prediction without GAR reaches 18.3%, while the complete predictive GAR model reaches 43.3%. Counterfactual successor prediction adds a further gain beyond order supervision. Raw successor distance is much weaker than the learned GAR energy in a one-seed local audit.

Controlled-future equivalence defines which language histories may share a state. Exact transition prediction leaves the geometry of inequivalent states ambiguous, whereas correct local action ordering is sufficient for shortest-path planning. GAR targets that order structure. A matched direct scorer, gradient-routing controls, and local geometry audits test whether the benefit enters through predictive geometry, supervised readout, or both.

2 PREDICTIVE STATES AND PLANNING GEOMETRY

Language-mediated control. Language provides many surface forms for the same situation. “The number of red objects is six” and “There are six red objects” may support exactly the same future calculations. Replacing either sentence by “There are not six red objects” can change every valid continuation. A useful state representation should ignore the first wording change and preserve the second semantic change.

We model this setting as a controlled process

$$\mathcal{M} = (\mathcal{S}, \mathcal{U}, P, \Omega, O, c, \mathcal{G}). \quad (1)$$

The semantic state is $s_t \in \mathcal{S}$ and a semantic intervention is $u_t \in \mathcal{U}$. The agent receives a language observation $x_t \in \Omega$ and chooses a language action a_t . A grounding map $\alpha(a_t) = u_t$ connects the

phrase to its effect. Decisions may depend on the full history

$$h_t = (x_0, a_0, x_1, \dots, a_{t-1}, x_t), \quad (2)$$

rather than on the latest sentence alone. Histories accommodate synonymous descriptions, irrelevant facts, and partial observations without requiring a network to recover a privileged hidden coordinate system.

Predictive equivalence. Predictive state representations define state through action-conditioned predictions of future observations (Littman et al., 2001). Let Y collect future task-relevant observations and terminal outcomes, let C collect future costs, and let F collect future feasible-action sets. Histories h and h' are controlled-predictively equivalent when every policy over actions admissible from both histories induces the same future law:

$$h \sim_{\text{pred}} h' \iff \mathcal{L}_\pi(Y_{t+1:\tau}, C_{t:\tau}, F_{t:\tau} \mid h) = \mathcal{L}_\pi(Y_{t+1:\tau}, C_{t:\tau}, F_{t:\tau} \mid h') \quad \forall \pi. \quad (3)$$

The corresponding predictive state is the equivalence class

$$q(h) = [h]_{\sim_{\text{pred}}} \in \mathcal{H}/\sim_{\text{pred}}. \quad (4)$$

Histories with identical controlled futures meet in the quotient even when their wording differs. A negation, operator substitution, or changed entity relation remains visible whenever it changes a future consequence. Information that cannot affect any future task outcome may disappear.

Proposition 1 (Control sufficiency). *If $h \sim_{\text{pred}} h'$ includes equality of future costs and terminal events under every admissible policy, then $V^\pi(h) = V^\pi(h')$ for every policy π , and therefore $V^*(h) = V^*(h')$.*

Proof. Return is a function of the future cost and terminal sequence. Equation 3 assigns that sequence the same distribution under each policy. Its expectation is therefore equal. Maximizing over policies preserves the equality. $\square$

Proposition 1 defines the state required for control. We do not claim it as a new control theorem. Bisimulation gives a related recursive characterization through agreement in immediate reward and transitions between equivalence classes (Ferns et al., 2011). GAR imposes a narrower condition, namely goal-conditioned order among actions available at one history.

The experiments instantiate the framework with a deterministic finite-horizon process. Executing a appends one deterministic textual consequence, written $T(h, a)$. Stochastic environments would require a predictive distribution or another representation of multimodal futures. A point predictor trained by mean-squared error generally represents only a conditional mean.

Coordinate ambiguity. Suppose an encoder ϕ and predictor F recover every deterministic transition:

$$F(\phi(h), e(a)) = \phi(T(h, a)). \quad (5)$$

Proposition 2 (Predictive ambiguity). *Let b be a bijection on the realized latent states and define $\tilde{\phi} = b \circ \phi$ together with*

$$\tilde{F}(b(z), e(a)) = b(F(z, e(a))). \quad (6)$$

Then $(\tilde{\phi}, \tilde{F})$ predicts the same realized transitions exactly. A non-isometric b need not preserve the order of successor distances to a goal.

Proof. Substitution gives $\tilde{F}(\tilde{\phi}(h), e(a)) = \tilde{\phi}(T(h, a))$. On a finite realized state set, the images of two successors and a goal may be placed at arbitrary distinct points. Their goal-distance order can change without changing any transition identity. $\square$

The two embeddings in Figure 1 realize the same transitions and disagree on the preferred successor. Additional assumptions can reduce this ambiguity. For example, controlled Gaussian world models with spectral separation and sufficient conditional action excitation can be identified up to an orthogonal map (Zhang et al., 2026). Proposition 2 concerns predictive learning without such metric-preserving guarantees.

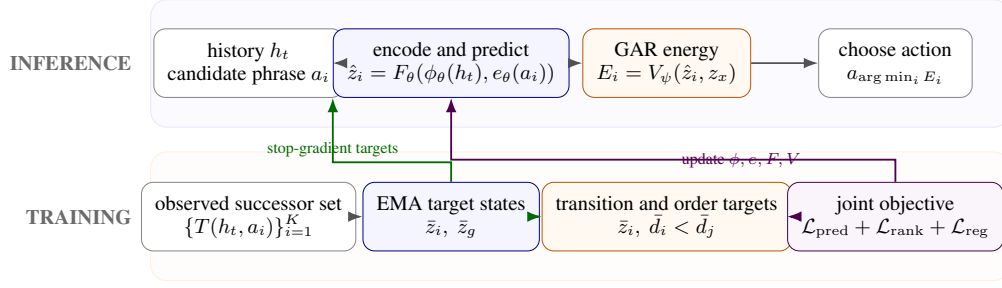


Figure 2: Inference and training paths for predictive GAR. At inference, each candidate phrase is encoded, passed through the successor predictor, and scored against the problem anchor. Training observes factual and counterfactual successor histories. EMA target distances define within-state order labels, while latent prediction preserves the action-conditioned transition.

Predictive architecture. The online encoder produces $z_h = \phi_\theta(h)$ and the action encoder produces $e_\theta(a)$. Their predicted successor is

$$\hat{z}_{h,a} = F_\theta(z_h, e_\theta(a)). \quad (7)$$

An exponential-moving-average encoder maps each observed successor history to a stopped-gradient target $\bar{z}_{T(h,a)}$. Factual and counterfactual prediction losses match $\hat{z}_{h,a}$ to these targets. The target encoder stays in evaluation mode and every dropout rate is zero.

At inference time, the prompt and query are represented by a problem anchor z_x . The transition-value realization assigns a lower-is-better energy

$$E_\psi^{\text{TV}}(h, a, x) = V_\psi(\hat{z}_{h,a}, z_x). \quad (8)$$

The policy predicts and scores every currently feasible phrase, then selects the minimum-energy candidate. Figure 2 separates this inference path from the training signals.

Geometric Advantage Ranking. Let $\bar{z}_g$ be the EMA representation of the terminal training history. Each observed candidate successor receives the target distance

$$\bar{d}_i = d(\bar{z}_{T(h,a_i)}, \bar{z}_g). \quad (9)$$

The default target evaluates a candidate after a bounded two-step reference continuation. It remains a geometric progress surrogate rather than a Bellman-consistent advantage. Whether its order agrees with true continuation quality is measured directly in the action audits.

Whenever $\bar{d}_i + \epsilon < \bar{d}_j$, GAR imposes

$$\mathcal{L}_{\text{rank}} = \sum_{i>j} [m + E_\psi(h, a_i, x) - E_\psi(h, a_j, x)]_+. \quad (10)$$

A calibration term matches differences in predicted energy to differences in target distance:

$$\mathcal{L}_{\text{reg}} = \sum_{i<j} [(E_j - E_i) - \beta(\bar{d}_j - \bar{d}_i)]^2. \quad (11)$$

Together with factual and counterfactual latent prediction, the training objective is

$$\mathcal{L} = \mathcal{L}_{\text{factual}} + \lambda_{\text{cf}} \mathcal{L}_{\text{cf}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}. \quad (12)$$

GAR gradients reach the scorer, predictor, action encoder, and online history encoder in the default model. The EMA network follows the online encoder without receiving gradients.

GAR names the supervision, not the scorer architecture. Transition-value GAR factors the energy through $\hat{z}_{h,a}$. Direct GAR uses $D_\psi(z_h, e(a), z_x)$ and bypasses the successor in the policy path. Raw-distance GAR uses $d(\hat{z}_{h,a}, \bar{z}_g)$ without a learned head. The last variant receives the terminal training representation during evaluation and is reported only as an oracle-terminal diagnostic.

Local planning condition. Let $D^*(h)$ denote the shortest remaining action count in a deterministic unit-cost puzzle, and define

$$A^*(h, a) = D^*(h) - 1 - D^*(T(h, a)). \quad (13)$$

An action on a shortest solution has $A^* = 0$. An action that spends a step without reducing the remaining distance has $A^* = -1$.

Proposition 3 (Local ordering). *If a learned score ranks every optimal action above every suboptimal action at every reachable history, greedy selection reaches the goal in exactly $D^*(h_0)$ actions.*

Proof. At remaining distance $d > 0$, the selected action reaches a history with distance $d - 1$. Repeating the argument reaches distance zero after d actions. $\square$

If the smallest true action gap is $\gamma > 0$, a uniform estimation error below $\gamma/2$ preserves the same ordering. GAR does not receive A^* during main training. The proposition instead identifies the behavioral condition that its geometric labels must recover. Pairwise ordering, top-1 recall, regret, and margin therefore provide more direct tests than effective rank or global probe accuracy.

Compression. The quotient is the coarsest exact partition that preserves controlled futures. It supplies a formal notion of task-relevant compression, but the neural encoder is not guaranteed to find that minimum. The information bottleneck likewise seeks a compressed code that preserves target information (Tishby et al., 2000). Our objective contains no rate penalty, so latent width and effective rank cannot be interpreted as evidence for an explicit information bottleneck.

3 EXPERIMENTAL DESIGN

Domain and interface. Stylized iGSM generates arithmetic dependency problems with named quantities, distractor relations, and a target query (Ye et al., 2025). Training instances are generated online. The paper models receive 300,000 examples with shortest solutions ranging from three to nine actions.

At each decision point, the environment supplies a shuffled set of executable intent phrases. It reveals neither the useful action nor the outcomes of unchosen candidates. All learned policies receive the same set and observe the textual consequence of their choice before acting again. The supplied menu is candidate-privileged relative to unrestricted generation. Holding it fixed isolates consequence evaluation from proposal recall and syntax validity.

Models. The main JEPA uses 256-dimensional history and action representations and an MLP successor predictor. A second JEPA uses a causal Transformer over encoded history states. The token LM scores candidate phrases by token likelihood. The sentence LM predicts the next intent phrase as a sentence-level target. The sentence-plus-latent model adds next-sentence latent prediction to the sentence objective. Each policy selects from the same current menu.

The main GAR model uses the transition-value factorization from Section 2. Objective controls remove ranking, remove pairwise calibration, or remove counterfactual latent prediction. Architecture controls detach GAR gradients from the predictive body, replace the learned energy with raw successor distance, or apply GAR to a direct scorer that bypasses the successor in the policy path. A shuffled-label control retains the candidate sets and counterfactual transitions while destroying their within-state quality correspondence.

Training and selection. Every completed learned headline row and primary ablation uses five training seeds. Hyperparameters are selected on validation data. Each model-dataset pair receives its own learning-rate search budget. Ablations reuse the selected rate unless their loss scale or gradient route changes, in which case they receive a local learning-rate check. Test results never select hyperparameters.

Evaluation. Strict success requires the shortest possible solution. Success at +2 allows at most two excess actions. Each completed seed is evaluated on 200 newly generated episodes. Tables report the mean and sample standard deviation across seed-level success rates. Episodes were not paired across methods, so the current estimates do not support paired model-difference claims.

Table 1: Closed-loop planning on stylized iGSM. Values are mean $\pm$ sample standard deviation over five seeds. Every method scores the same shuffled current feasible-action menu.

Model	Strict success	Success with +2 actions
Sentence LM + latent	.521 $\pm$.047	.809 $\pm$.020
Sentence LM	.471 $\pm$.046	.813 $\pm$.023
MLP JEPA + GAR	.433 $\pm$.019	.796 $\pm$.033
Token LM	.422 $\pm$.124	.836 $\pm$.073
Causal JEPA + GAR	.366 $\pm$.047	.778 $\pm$.043

Table 2: Five-seed objective ablations. Rank is the GAR pairwise hinge. Reg. calibrates energy differences to geometric target differences. CF pred. predicts counterfactual successor states.

Configuration	Rank	Reg.	CF pred.	Strict	+2
Full predictive GAR	yes	yes	yes	.433 $\pm$.019	.796 $\pm$.033
GAR ranking only	yes	no	yes	.358 $\pm$.045	.750 $\pm$.018
GAR without CF prediction	yes	yes	no	.345 $\pm$.049	.734 $\pm$.017
Latent prediction only	no	no	yes	.183 $\pm$.016	.628 $\pm$.031
Factual prediction only	no	no	no	.076 $\pm$.013	.422 $\pm$.022

Diagnostics. The local decision audit enumerates the same feasible actions at aligned histories. It records pairwise ranking accuracy, top-1 optimal-action recall, mean regret, rank correlation with exact continuation cost, and the optimal-versus-best-suboptimal margin. Exact continuation cost is an evaluation label and is never available to the policy.

Frozen-feature probes measure operation identity, remaining actions, resolved quantities, and numerical outcomes. Effective rank checks for collapse. The current feature set does not contain controlled paraphrase, negation, operator-swap, or entity-renaming pairs, so no claim about semantic invariance is made.

Scope. Only stylized iGSM currently has a complete five-seed headline comparison. Faithful iGSM and non-arithmetic domains remain outside the main claims. Test-time scaling is also excluded. The current evaluator and FLOP accounting do not yet isolate the effect of added inference computation.

4 PLANNING RESULTS

Model comparison. Table 1 addresses RQ1. MLP JEPA with GAR attains .433 strict success. Its mean is close to the token LM at .422, below the sentence LM at .471, and below the sentence-plus-latent model at .521. Causal JEPA attains .366. MLP JEPA has the smallest standard deviation across the five tested seeds, although the sample is too small to support a general stability claim.

Non-generative latent consequence prediction supports closed-loop planning under the common interface, but it does not outperform the sentence models. The leading sentence-plus-latent row is consistent with complementary benefits from generative and predictive objectives.

Objective ablation. Table 2 addresses RQ2. Factual and counterfactual latent prediction without GAR attains .183 strict success. Predictive GAR attains .433. Factual prediction without GAR or counterfactual prediction attains .076. Ranking without pairwise calibration attains .358. GAR without the counterfactual latent loss attains .345.

Same-state action ordering accounts for the larger part of the improvement. Counterfactual successor prediction supplies an additional gain beyond the order labels. Merely exposing the predictor to factual transitions is insufficient.

The reference horizon produces a sharply non-monotonic result. Horizons one, two, and four attain .161, .433, and .181 strict success. Horizon two is best among the tested settings, but the experiment does not support a universal optimum. Increasing counterfactual breadth from one to four changes

Table 3: Five-seed frozen-feature diagnostics. Probe scores are means over seeds. Effective rank measures non-collapse and does not track strict planning performance.

Model	Eff. rank	Operation bal. acc.	Remaining R^2	Resolved R^2	Outcome R^2
MLP JEPA	198.70	.406	.346	.895	.088
Causal JEPA	215.34	.403	.332	.846	.053
Token LM	3.55	.532	.439	.986	-.020
Sentence LM	79.64	.777	.410	.879	-.027
Sentence LM + latent	71.44	.807	.429	.886	-.029

strict success from .403 to .439. Pairwise-regression weights .10 and .50 attain .443 and .402, so the gain persists across nearby coefficients.

Information controls. Shuffling geometric quality labels while retaining counterfactual transitions gives .015 strict success at each of three screened learning rates. Direct GAR without successor auxiliaries reaches .150 to .185. Adding the JEPA successor objectives to the same direct policy path raises one-seed performance to .440, compared with .435 for transition-value GAR in that seed.

Counterfactual volume alone cannot explain the GAR result. Predictive auxiliary training clearly helps a direct scorer, but the current in-distribution comparison does not show that the policy itself must factor through a predicted successor. The stronger factorization claim requires an advantage in data efficiency, compositional length, semantic robustness, goal transfer, recovery after errors, or valid multi-step simulation.

5 REPRESENTATION ANALYSIS

Local action order. RQ3 concerns within-state action ordering rather than global cluster structure. In a one-seed aligned-history audit, the deployed transition-value energy attains .826 pairwise accuracy and .786 top-1 optimal-action recall. Raw predicted successor distance attains .603 and .549. The learned energy orders actions far better than uncalibrated latent distance in that checkpoint.

The audit does not yet isolate representation shaping from supervised readout. Full predictive GAR reaches .435 strict success in the matched seed. Detaching GAR gradients from the predictive body reaches .375, raw-distance selection reaches .190, and direct GAR with successor auxiliaries reaches .440. Five-seed completion and checkpoint-level local metrics are needed before assigning the full gap to changes in the predictive representation.

Because the audit covers one seed, it does not establish whether local ordering tracks planning across checkpoints or runs. It also does not establish that GAR reshapes the predictor rather than training a stronger readout.

Global decodability. Table 3 reports the completed frozen-feature diagnostics for RQ4. Both JEPA spaces are non-collapsed, with effective ranks near 200 in 256 dimensions. Token-LM features have effective rank 3.55 while supporting competitive planning. Effective rank is therefore a health check rather than an explanation of control performance.

Sentence representations make operation identity substantially more linearly accessible than JEPA representations. MLP JEPA strongly exposes resolved-variable count, but resolved count is confounded with elapsed step index. Remaining distance is moderately decodable across all systems. Numerical outcomes are weakly decoded. The existing necessary-action probe is at chance and provides no positive evidence.

The quotient account also predicts invariance to surface paraphrase and sensitivity to meaning-changing edits. The available probes do not test those predictions. We therefore omit t-SNE and UMAP projections, which would be difficult to interpret without controlled semantic pairs and corresponding full-space statistics.

6 RELATED WORK

Predictive state and control equivalence. Predictive state representations describe state through multi-step action-conditioned predictions of future observations (Littman et al., 2001). Bisimulation identifies states that agree in reward and transition behavior, while approximate bisimulation metrics relate representation distance to value continuity (Ferns et al., 2011). We use these ideas to define the control-sufficient quotient over language histories. We do not claim to learn the complete quotient or a full bisimulation metric.

Joint-embedding prediction. JEPA proposes learning predictive representations without reconstructing all observation detail (LeCun, 2022). I-JEPA and V-JEPA instantiate feature prediction for images and video (Assran et al., 2023; Bardes et al., 2024). V-JEPA 2 adds an action-conditioned model for robot planning toward image goals (Assran et al., 2025). Language-focused variants include LLM-JEPA and JEPA-Reasoner (Huang et al., 2026; Liu et al., 2026). Our subject is controlled language-action consequence prediction and same-state counterfactual action ordering.

Planning-aware predictive geometry. Recent methods also address the gap between predictive representations and planning. Value-guided JEPA trains goal-conditioned distance with a value-learning objective (Destrade et al., 2026). Temporal straightening regularizes latent trajectory curvature (Wang et al., 2026). Temporal-Distance JEPA learns a directed temporal cost from trajectory order, cross-trajectory negatives, and rollout consistency (Bai & Xiong, 2026). GAR instead imposes within-state order constraints over alternative discrete language actions. Its transition-value realization distills that ordering into a scorer over predicted successors. Direct GAR tests whether the predictive factorization adds value beyond the same supervision.

The overlap rules out priority claims for JEPA planning or planning-aware latent geometry. Our contribution is the action-level counterfactual formulation, its control-sufficiency account, and a controlled comparison of predictive and direct candidate scoring in language.

Latent world models and shaping. Dreamer and MuZero show that latent dynamics become decision useful when policy, reward, or value objectives shape their representations (Hafner et al., 2020; Schrittwieser et al., 2020). Potential-based shaping adds a fixed potential difference while preserving optimal policies under stated conditions (Ng et al., 1999). GAR learns its representation, geometric targets, and energy jointly, so it is related to shaping but not an instance of the classical fixed-potential result.

Information bottleneck and test-time computation. The information bottleneck formalizes compression that preserves target-relevant information (Tishby et al., 2000). We use it only as a related desideratum because the present model has no explicit rate penalty. More inference computation can improve reasoning through search, sampling, or recurrent depth (Snell et al., 2024; Geiping et al., 2025). We do not report a FLOP-matched scaling result.

7 CONCLUSION

Controlled-future equivalence specifies which language histories may share a planning state. Action-conditioned prediction models transitions between such states, but does not determine how their consequences should be ordered. GAR supplies that local order supervision.

On stylized iGSM, predictive GAR turns a weak latent-prediction policy into a viable conditional planner. Counterfactual ordering contributes the larger gain, and successor prediction adds a further improvement. The strongest sentence model remains better in strict success. Raw distance also remains much weaker than the learned GAR energy.

Direct GAR matches the predictive policy in the current one-seed comparison. The experiments therefore do not yet show that a policy must factor through its predicted successor. They support the narrower claim that successor prediction supplies useful auxiliary training under matched order supervision.

The present evidence concerns conditional planning under a supplied feasible-action menu on a synthetic arithmetic domain. It does not yet support claims about unrestricted language agents or general-purpose reasoning.

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A LIMITATIONS AND SCOPE

The main interface supplies the current symbolic feasible-action menu, isolating action selection from unrestricted proposal and language generation. The reported task is conditional planning among executable phrases rather than autonomous language agency.

The completed evidence comes from a stylized arithmetic environment. Templated language and exact counterfactual transitions enable mechanism analysis, but they limit external validity. No result establishes transfer to faithful iGSM or a non-arithmetic domain.

Several mechanism controls use one seed. Direct GAR matches transition-value GAR in that setting. Detached gradients, raw-distance planning, and the local metric audit therefore remain exploratory and do not support population-level comparisons.

The deeper-planning protocol does not support a test-time-scaling claim because proposal work, recursive prediction error, and recurrent-baseline FLOPs are not yet jointly controlled. No scaling curve is reported.

The present model is trained from scratch on a controlled language. Its findings do not directly establish how pretrained large language models would organize open-domain planning states. Conversely, the controlled setting enables exact causal comparisons that would be difficult to interpret in an open-ended agent benchmark.

B DETAILED PROTOCOL

Information boundary. Headline policies observe the problem statement, prior selected intent phrases and outcomes, and a shuffled menu of actions executable at the current state. They do not observe the expert action, exact remaining distance, future action availability, terminal state, or counterfactual outcomes at evaluation. Training uses exact alternative outcomes only for declared counterfactual prediction and GAR targets.

Target encoder. The EMA encoder is updated from the online encoder and kept in evaluation mode at all times. Gradients do not enter EMA targets. Dropout is disabled in the encoder, predictor, action encoder, scorer, and target encoder. Earlier validity tests found that stochastic or train-mode targets destabilized dense rollout supervision.

Evaluation samples. The completed five-seed results use 200 fresh episodes per seed. Generated puzzle indices and candidate-order seeds were not paired across methods. We therefore report sample standard deviations across seed-level rates and do not report paired confidence intervals or paired model differences.

Tuning contract. Width-256 learning rates are screened on validation and reused within a model-dataset pair. The grid is $\{5e-5, 7e-5, 10^{-4}, 3e-4, 5e-4, 7e-4, 10^{-3}, 3e-3, 5e-3\}$. Five seeds are trained after selecting the validation learning rate. Ablations reuse that rate unless they change gradient routing or loss scale.

Historical comparability. Historical reports used different generators, predictor variants, batch sizes, candidate interfaces, and training budgets. The manuscript includes only the 300,000-example paper family and matched controls retrieved from its run lineage. Older values remain in the repository but are not combined with the current table.

C COMPLETE CURRENT RESULT TABLES

Table 4: Five-seed GAR ablations on stylized iGSM.

Variant	Strict	+2 actions
Full GAR	.433 $\pm$.019	.796 $\pm$.033
No GAR	.076 $\pm$.013	.422 $\pm$.022
Latent MSE only	.183 $\pm$.016	.628 $\pm$.031
Ranking only	.358 $\pm$.045	.750 $\pm$.018
No counterfactual latent loss	.345 $\pm$.049	.734 $\pm$.017
GAR horizon 1	.161 $\pm$.021	.597 $\pm$.037
GAR horizon 2	.433 $\pm$.019	.796 $\pm$.033
GAR horizon 4	.181 $\pm$.008	.651 $\pm$.014
One counterfactual	.403 $\pm$.041	.772 $\pm$.020
Four counterfactuals	.439 $\pm$.049	.821 $\pm$.025
Advantage MSE weight .10	.443 $\pm$.037	.802 $\pm$.034
Advantage MSE weight .50	.402 $\pm$.039	.783 $\pm$.026

Table 5: Exploratory one-seed mechanism controls. Raw distance uses an oracle-terminal embedding and is candidate privileged. GAR names the supervision. TV and direct name the scoring factorization.

Variant	Strict	+2 actions
Transition-value GAR	.435	.820
Detached GAR body	.375	.775
GAR without counterfactual latent loss	.255	.695
Counterfactual latent only	.080	.410
Direct GAR with successor auxiliaries	.440	.820
Raw successor distance [†]	.190	.625

Table 6: Five-seed representation diagnostics. Probe values are means across seeds.

Model	Effective rank	Operation bal. acc.	Remaining R^2	Resolved R^2	Outcome R^2
MLP JEPa	198.70	.406	.346	.895	.088
Causal JEPa	215.34	.403	.332	.846	.053
Token LM	3.55	.532	.439	.986	-.020
Sentence LM	79.64	.777	.410	.879	-.027
Sentence LM + latent	71.44	.807	.429	.886	-.029

D REPRODUCIBILITY AND ARTIFACT STATUS

The conceptual figures are repository-native TikZ and compile directly with the manuscript. Empirical results are transcribed from the retrieved current-findings ledger and linked to run identifiers in the paper README. No t-SNE, UMAP, or test-time scaling figure appears before its analysis protocol is valid and its source features are retained.

The completed runs use immutable code snapshots and checkpoint hashes. Failed and superseded jobs remain visible in the controller ledger. The manuscript excludes the current scaling pilots from

Table 7: One-seed information controls across a local learning-rate screen, separating meaningful counterfactual ordering from counterfactual exposure alone.

Control	Learning rate	Strict	+2 actions
Direct GAR without successor auxiliaries	10^{-4}	.180	.590
Direct GAR without successor auxiliaries	3×10^{-4}	.185	.635
Direct GAR without successor auxiliaries	7×10^{-4}	.150	.560
GAR with shuffled quality labels	10^{-4}	.015	.170
GAR with shuffled quality labels	3×10^{-4}	.015	.175
GAR with shuffled quality labels	7×10^{-4}	.015	.175

evidence. The final artifact will provide a machine-readable manifest with checkpoint path, hash, model configuration, seed, validation split, evaluation indices, parameter count, optimizer updates, wall time, and measured FLOPs for every included row.

The official ICLR 2027 style files are vendored unchanged in the paper directory. Run `make figures` through `make_figures.sh`, then compile `main.tex` with `latexmk -pdf`. The draft remains anonymous because `\iclrfinalcopy` is disabled.